

An algebraic power-tower value at an interior algebraic base

CONJECTURE 00000000217 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Interior base

The algebraic base $x=1$ lies strictly between $\exp(-e)$ and $\exp(1/e)$. Every finite power tower at that base is 1, and the convergent tower value is 1.

Classification contradiction

The value 1 is algebraic, a root of $X-1$. Mahler S-numbers are transcendental; algebraic numbers belong to the A class. The claim that every interior algebraic base gives an S-number therefore fails.

Scope

This refutes the interior-value conjunct only. No claim is made about the proposed lower-endpoint degeneration.

Formalization

Lean proves the strict interval inequalities, convergence of the actual finite tower sequence to one, algebraicity of the base and non-transcendence of the value. The written argument uses only the defining requirement that an S-number be transcendental.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000217.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.